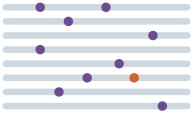
LIBRARIES
nanopore, Illumina or Sanger

AID-competent animals



AID-null controls (two or more)





nanopore Illumina Sanger

Sample sheet

file	animal	control
a.fastq	Wt 2	no
b.fastq	KO 1	yes
c12.ab1	Wt 3	no

labels stay with every flag

READ PREPARATION
every read is gated

Gates



too long, or > 1% errors/base
Sanger: Q ≥ 20 core kept

Competitive alignment



each read goes to its best match; a paralogue can be a target of its own

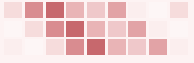
Read accounting



each drop with its reason, every library (Figure S1)

AUTOMATED ANALYSIS
calibrated by the AID-null controls

Background



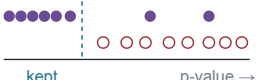
per position, change, strand and quality,
+ a leave-one-out margin

Candidate test



Poisson tail against the background; in ≥ 50% of reads: germline, set aside

Threshold set by the controls (target–decoy)



calls (filled) against decoys: each control called against the others (open)

threshold: predicted false calls ≤ 5% of calls kept

Sanger: Q ≥ 20 changes counted; background shown beside

Bookkeeping checks (reported, never applied)



copied trace | clone in two animals | mutated control | label mismatch | load outlier

RESULTS
per animal, then per group

Mutation profiles



Group tests

Welch t on $\log_{10}$ density, one value per animal



● mutation ● sequencing error